Subject-independent forearm sEMG electrode reduction with cross-fitted attribution and non-inferiority testing

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Formatting preflight: Introduction and Methods only. Experiments continue.

1 Introduction

Surface electromyography (sEMG) provides a non-invasive signal for gesture recognition and myoelectric control, but the recording interface must balance spatial information against the burden of additional electrodes (Scheme and Englehart, 2011; Farina et al., 2014; Vujaklija et al., 2016). Fewer recording sites could simplify an interface. Whether that reduction preserves recognition for an unseen participant is an empirical question, and offline recognition alone does not establish a usable control interface.

Channel selection has been studied with feature filters, wrappers and spatial criteria (Phinyomark et al., 2012; Stango et al., 2015; Qu et al., 2021). SHAP-guided EMG channel and feature optimization is also established (Rani et al., 2025); related work uses Shapley explanations for similar sEMG gestures (Wang et al., 2024) and explainable feature selection with fusion (Gehlot et al., 2024). These contributions make attribution-based reduction a relevant comparator rather than a new algorithmic premise. Their performance estimates cannot be directly compared with this study across different tasks, sensors and validation protocols.

For a system intended to work across people, selection must exclude the eventual test participant. Using that participant to standardize features, estimate importance or choose a subset can produce optimistic estimates. Participant-disjoint inner evaluation also avoids measuring attribution solely on observations used to fit the attributing model. This motivates an outer leave-one-subject-out (LOSO) evaluation in which each entire selection procedure is reconstructed from the outer training participants.

Attribution magnitude describes predictive reliance of a fitted model, not a direct measurement of muscle activation. TreeSHAP distributes a model's prediction over its input features (Lundberg and Lee, 2017; Lundberg et al., 2020); aggregation over an electrode's features offers a ranking, but does not turn it into a coalition-level Shapley value for that electrode. Correlated features and variation across participants can affect this ranking. Attribution stability has already motivated resampling and regularization methods (Visani et al., 2022; Xiang et al., 2026); a stability penalty therefore requires validation rather than a novelty claim.

Electrode count requires an explicit tolerance for performance loss. Non-inferiority asks whether a reduced system remains within such a tolerance, whereas a non-significant superiority test does not establish preservation (Schuirmann, 1987; Walker and Nowacki, 2011). A mean tolerance is also different from preserving every participant or gesture. Reporting only an aggregate curve could hide losses that matter to an intended application.

We evaluate cross-fitted electrode reduction on GRABMyo and separately examine NinaPro DB5. The primary question is the minimum count supported by an internally documented, predefined engineering margin. Secondary questions concern utility beyond random and spatial selection, stability regularization, and deletion with retraining. New exploratory checks use a geometric subset under the primary training protocol, paired participant and gesture predictions, and shared-resource SHAP and grouped permutation rankings. Sensitivities and split-derived layouts delimit the claim: the LOSO experiment evaluates a selection procedure, not a prospectively validated common physical layout.

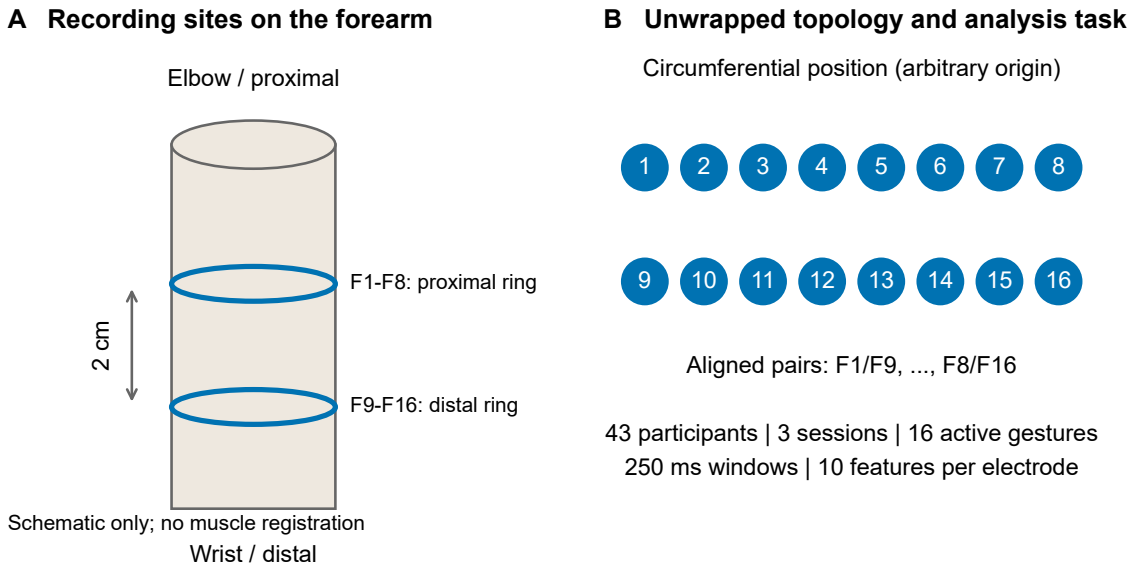


Figure 1. Recording topology and task. The source montage places F1–F8 in the proximal ring and F9–F16 in the distal ring, separated by 2 cm. Corresponding circumferential positions are aligned. The forearm outline is schematic; recording sites were not registered to individual muscles. The unwrapped map preserves electrode numbers without implying a verified absolute anatomical angle.

2 Methods

2.1 Datasets, selection units and signal processing

GRABMyo v1.1.0 contains 43 able-bodied participants recorded over 3 days with an EMGUSB2+ amplifier at 2048 Hz (Pradhan et al., 2022; Jiang et al., 2024). The forearm montage has sixteen monopolar sites in two rings (figure 1); each site supplies one channel, so electrode and channel selection coincide here. The primary task uses sixteen active gestures, excluding rest. All 30,715 distributed files matched published checksums, and 15,351 records were processed. Source demographics and the official gesture dictionary are provided with the reproducibility package.

Signals were filtered with a fourth-order zero-phase Butterworth band-pass at 20–450 Hz and a 50 Hz notch ($Q = 30$), then divided into 250 ms windows with 50% overlap within trial boundaries. The non-causal filter limits the analysis to offline use. Each channel supplied mean absolute value, RMS, variance, waveform length, zero crossings, slope sign changes, integrated EMG, Willison amplitude, mean frequency and median frequency. The active-task cache contains 563,472 windows with 160 features each.

Count-feature thresholds followed a common rule: 1.15 times the median absolute signal first difference, rounded to one significant figure. Historically, the monopolar threshold (10^{-2} mV) was derived from 54 unfiltered records from six participants before LOSO. It was not estimated separately within each outer fold. Subsequent exclusion checks reproduced the identical rounded threshold in all 43 folds. Thus the stored features are invariant to this particular exclusion, but the original development sequence should not be described as fully isolated. Fold-specific feature means and standard deviations, classifier fitting and attribution ranking exclude the outer test participant.

NinaPro DB5 was analyzed separately: 10 participants, two Myo armbands, sixteen channels sampled at 200 Hz and exercise 1's twelve finger movements (Pizzolato et al., 2017). Refined labels determine movement segments. Filtering was 10–95 Hz without a notch; 250 ms windows contain 50 samples. The threshold is seven quantization steps in the stored integer representation; every participant-exclusion check reproduced it. No anatomical correspondence between DB5 and GRABMyo channel numbers is assumed.

Ethics This is a secondary analysis of publicly distributed, de-identified recordings, with no new recruitment or recording. Original study approvals and consent are described in the dataset publications. [AUTHOR INPUT REQUIRED: institutional determination or exemption for this secondary analysis remains unresolved.]

2.2 Outer evaluation and classifier budgets

Each outer LOSO fold reserves all observations of one participant p. An inner five-fold GroupKFold partitions only the other participants (figure 2). Inner feature standardization uses inner-training observations only; final reduced models use the corresponding outer-training statistics. No outer test observations enter fitting or rankings.

The primary reduction curve uses XGBoost with 300 trees, maximum depth 6, learning rate 0.1, row and column subsampling 0.8, histogram tree construction and fixed seed 20260728. Its multiclass soft-probability objective produces probabilities whose argmax gives the predicted class. Every count and ranking within an outer fold uses the same stratified sample of 200,000 training windows. Models are retrained from scratch on the selected features.

Training budgets differ across historical tracks: the full-array XGBoost classifier context uses all training windows, the primary curve uses 200,000, and the random-selection comparison uses 50 trees and 150,000. The full-array SVM uses 40,000 windows; random forest uses 200,000. These are budget-limited comparators, not proven lower bounds or matched classifier-capacity experiments. Budgets were calibrated using two outer-fold outcomes. The internally dated analysis plan followed an earlier pilot; no independent registration or Git commit authenticating its date is available. These development exposures qualify claims of wholly prospective evaluation.

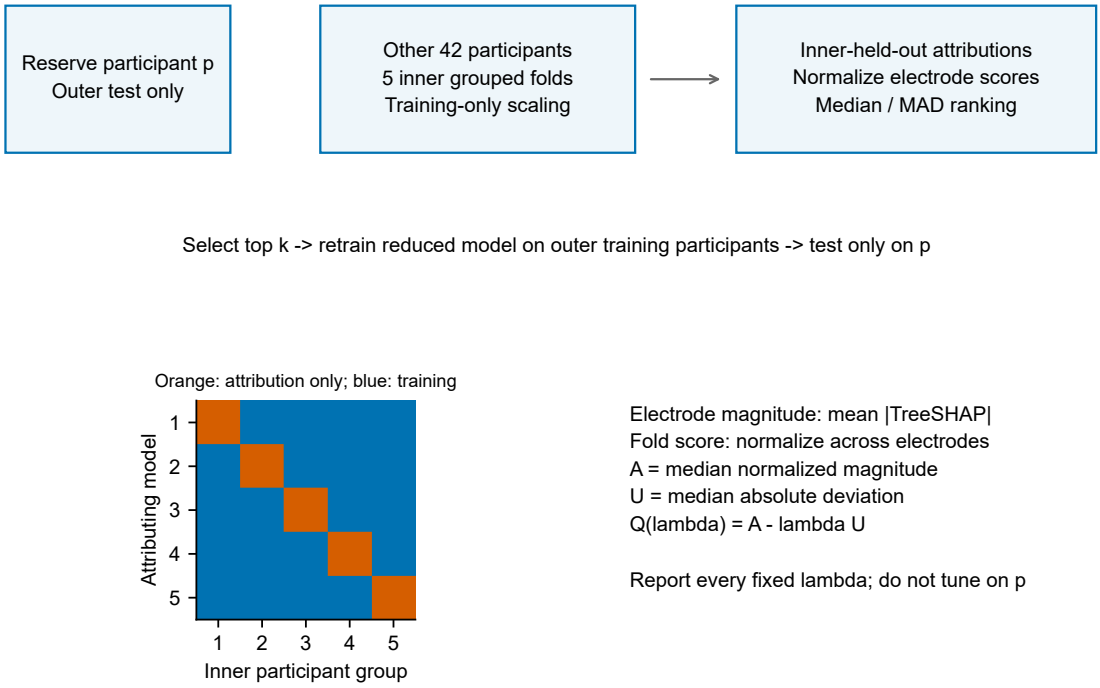


Figure 2. Participant access in the nested procedure. Each inner attributing model excludes the participants on whom its attributions are evaluated. The outer participant is reserved until final reduced-model testing. This diagram describes fold-specific scaling, fitting and ranking; historical threshold pooling and budget calibration are separate development exposures documented in the text.

2.3 Cross-fitted attribution and its participant-level extension

Inner XGBoost models use at most 200,000 sampled training windows. Exact tree-path-dependent TreeSHAP contributions are computed on raw class margins for at most 20,000 inner-held-out windows. For electrode c, absolute contributions are averaged across windows, classes g and its ten features f:

$$\Phi_{i,c} = \frac{1}{GF} \sum_{g=1}^{G} \sum_{f=1}^{F} |\phi_{i,g,c,f}|. \tag{1}$$

The window mean is normalized across electrodes to give the inner-fold score $I_{j,c}$. The central magnitude and unscaled median absolute deviation are

$$A_c = \text{median}_j \, I_{j,c}, \quad U_c = \text{median}_j \, |I_{j,c} - A_c|, \quad Q_c(\lambda) = A_c - \lambda U_c. \tag{2}$$

All fixed penalties $\lambda \in \{0, 0.25, 0.5, 1\}$ are reported, with lower electrode index breaking ties. No penalty is chosen using outer scores. The unpenalized ranking is called cross-fitted SHAP; the positive penalties assess stability regularization.

Pooling people within five inner folds can suppress person-level heterogeneity. A post-hoc extension therefore computes the median and MAD over normalized individual attribution vectors of all 42 outer-training participants. These vectors share five fitted inner models; their dispersion is descriptive heterogeneity, not a standard error. All penalties and counts are evaluated under the unchanged primary protocol. Exact matching participant/subset fits may be reused with source-row provenance; new subsets require new fits. A historical source-hash mismatch is preserved in the audit, and current reporting uses an isolated refit with current input fingerprints.

2.4 Primary endpoint and participant-level inference

Macro-F1 is the unweighted mean of the gesture F1 scores for each held-out participant. For count k, the paired change is $\Delta_{p,k} = \mathrm{F1}_{p,k} - \mathrm{F1}_{p,16}$. The predefined primary engineering tolerance is $\delta = 0.02$ on the 0–1 scale. The historical decision rule requires the lower 95% percentile-bootstrap bound on the participant mean change to exceed $-\delta$; k^* is the smallest qualifying count below sixteen. Sensitivity margins are 0.01 and 0.03. This criterion concerns mean recognition performance, has no established clinical interpretation and does not imply equivalence or individual preservation.

The participant is the statistical unit: 43 for GRABMyo, ten for DB5 and 21 or 22 test participants for split-layout checks. Bootstrap intervals use 10,000 resamples of paired participant scores with seed 20260728. They are conditional on the observed fitted LOSO models. Overlapping training cohorts and the lack of refitting within every bootstrap mean these intervals do not capture all new-cohort training uncertainty (Bengio and Grandvalet, 2004).

The original Holm procedure used the fraction of unshifted bootstrap mean differences at or below $-\delta$, with a Monte Carlo floor. This is a percentile-tail quantity rather than a bootstrap distribution generated under the null. We preserve it and additionally report an exploratory, null-centred studentized bootstrap (Davison and Hinkley, 1997). For $d_p = \Delta_{p,k}$, residuals $d_p - \bar{d}$ are resampled, their studentized means are compared with $(\bar{d}+\delta)/(s_d/\sqrt{n})$, and the upper-tail probability uses a plus-one correction. Holm adjustment covers the fifteen reduced-count tests at one-sided $\alpha = 0.025$, separately for each condition, margin and penalty. No seed or margin is chosen from results.

Paired secondary contrasts use Wilcoxon signed-rank tests and participant mean-change percentile intervals. These address different estimands: signed ranks are not a test of the mean. Holm families are stated explicitly; a non-significant contrast is not evidence of equivalence. The detailed historical secondary families were specified after their underlying results became available and are labeled secondary rather than prospectively confirmatory.

2.5 Selection comparators and attribution diagnostics

Historical lighter-model comparisons use random subsets, cross-fitted SHAP, stability-regularized SHAP ($\lambda = 0.5$), feature-wise permutation, mutual information and geometric farthest-point selection. Random selection uses 100 subsets per participant at counts 4, 6 and 8, and 500 at count 12. Eight attribution-versus-mean-random contrasts receive Holm correction. Historical permutation rankings use a 20,000-window fitted model, 8,000 validation windows and three feature-wise permutations, whereas SHAP uses the inner cross-fitting budget above. Thus shared final evaluation does not make their selection resources comparable.

New exploratory geometry12 testing uses the primary 300-tree, 200,000-window protocol, the same outer training sample and full16 reference. Geometric selection uses the fixed two-ring topology, nominal 24 cm circumference and 2 cm separation, starts at F1 and successively maximizes distance from selected sites; ties favor the lower index. It is not tuned to participant outcomes.

A separate exploratory resource-matched comparison shares five inner models between SHAP and grouped-electrode permutation, with up to 200,000 training and 2,000 validation windows per fold. Equal validation quotas represent every inner-held-out participant. All ten features of one electrode are permuted jointly within each participant, three repeats, scoring the equally weighted mean participant Macro-F1 loss. Signed losses are retained, normalized by their summed absolute magnitude, and aggregated by the median across inner folds. SHAP uses the identical observations and normalizes absolute contribution magnitudes. Final models at counts 6 and 12 use the primary protocol; two contrasts receive Holm correction. Measured training and selection times are reported separately. Because both grouping and validation resources differ from the historical permutation method, this experiment does not isolate budget alone.

Deletion with retraining compares removal of high-ranked, random and low-ranked electrodes using the lighter model. Its fifteen contrasts form one Holm family. This diagnostic tests predictive ranking relevance; it cannot identify causal muscles or validate an online controller.

2.6 Participant and gesture degradation

Primary full16, SHAP12 and geometry12 models were rerun to retain predictions indexed to the same canonical outer-test rows. Full16 and SHAP12 scores must reproduce the original curve within 10^{-6}. Participant losses, quantiles, worst loss and the number exceeding the primary margin are descriptive. Gesture-specific F1 differences are paired over all participants, with Holm correction across sixteen gestures. Confusion changes average row-normalized confusion matrices within participants so people, rather than windows, receive equal weight. Gesture names follow the explicit numeric ordering in MotionSequence.txt.

2.7 Sensitivity, transfer and fixed-layout scope

Rest-inclusive recognition and 150/500 ms windows use raw-signal re-extraction and complete count curves with the primary 250 ms ranking held fixed. The 150 ms window is 307 samples with stride 153; 250/500 ms use 512/1024 samples and half-window strides. The same 200,000-window count does not equate recording duration across windows. A random-forest sensitivity uses the pre-existing 200-tree, minimum-leaf-size-2 configuration at counts 12 and 16, the unchanged XGBoost ranking and the same training sample. It tests those counts only and cannot establish a random-forest minimum count.

DB5 applies the procedure separately with its own sixteen-channel reference. Derived bipolar signals $B_j = F_j - F_{j+8}$ use eight aligned pairs, re-extracted features and the eight-pair reference; they are not physical bipolar-sensor measurements. Full-array cross-session analyses compare training on session 1 with testing on sessions 1–3, and training on sessions 1–2 with testing on session 3. They do not test a reduced layout under displacement, and the latter comparison confounds data quantity with session diversity.

Fixed-layout checks use odd-ID development/even-ID testing and the reverse, each deriving a layout using development participants only. The count of twelve had already been selected using the full cohort, so neither direction independently validates count selection. The layouts differ and must be reported separately. No fresh, independently collected cohort with this montage was available. Feature redundancy and ranking-agreement diagnostics are documented in the Supplementary and machine-readable outputs; high outer-fold agreement is qualified by extensive training-set overlap.

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